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NPS/003/022 – Technical Specification for 33kV Disconnectors, Switch-Disconnectors and Earth Switches

1. Purpose

This document is the technical specification for 33kV Plain Break Disconnectors, Switch-Disconnectors and Earth Switches utilised on the NEDL and YEDL networks.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
DSS/011/016	Rev 0	Specification for 33kV (36kV) Disconnectors, Switch Disconnectors and Earth Switches

2. Scope

This specification details the requirements for 33kV plain break disconnectors, switch disconnectors and earth switches for use at NEDL and YEDL open compound substations. It will be necessary to consider and include any project specific requirements as detailed in Appendix 3, Addendum to Supplier Requirements.

3. Technical Requirements

Disconnectors must comply with Energy Networks Association Technical Specification (ENATS) 41-36 plus those additional requirements listed in this document and its appendices. Disconnectors will preferably have an ENA Certificate of Conformance.

Appendix 1 of this document details the requirements selected within ENATS 41-36, or where the requirements of ENATS 41-36 have been supplemented.

Appendix 2 shall be completed to confirm the technical specification of each equipment type.

Appendix 3 contains an Addendum to Supplier Requirements which lists any project specific requirements.

4. References

4.1 External Documentation

Reference	Title
ENATS 41-36	Distribution Switchgear for Service up to 36KV (Cable and Overhead Conductor Connector)

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4.2 Internal documentation

Reference

Title

5. Definitions

Term

Definition

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
K.Midgley	Standards Engineer	K.Midgley	01/02/2006

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G.Bolden	Purchasing Manager	G.Bolden	09/02/2006

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
D.Gazda	Standards Team Co-ordinator	D.Gazda	09/02/2006

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S.Johnson	DBD Administrator	S.Johnson	24/03/2006

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C.Holdsworth	Standards and Assessment Manager	C.Holdsworth	13/02/2006

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C.Morgan	Head of Procurement	C.Morgan	20/02/2006

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Appendix 1: Technical specification supplementing ENATS 41-36

33kV Disconnecter

	ENATS 41-36 sub clause	33kV disconnecter	33kV switch- disconnecter
Particulars of system			
Voltage		33kV	33kV
Frequency		50Hz	50Hz
Number of phases		3	3
Neutral earthing		Solid	Solid
Switch characteristics			
Number of poles		3	3
Switch-disconnector category A-B	6.3		B
Indoor, outdoor	1.2	Outdoor	Outdoor
Rated voltage	1.4.1	36kV	36kV
Rated insulation level (lightning impulse withstand voltage)	1.4.2	170/195kV	170/195kV
Rated insulation level (1 minute power RMS power frequency)	1.4.2	70/80kV	70/80kV
Rated Frequency	1.4.3	50Hz	50Hz
Disconnecter / Switch-disconnector rated normal current	1.4.4	800A	800A
Rated short-time withstand current	1.4.5	25kA	25kA
Rated duration of short circuit	1.4.7	3 sec	3 sec
Rated supply voltage of control, closing and opening devices	1.4.8	110V	110V
Rated supply frequency of control, closing and opening and of auxiliary circuits.	1.4.9	DC	DC
Switch disconnector. Rated mainly active load breaking current.	6.4.101		400A
Switch disconnector. Rated closed-loop breaking current	6.4.102		400A
Switch disconnector. Rated cable charging breaking current	6.4.104		20A
Switch disconnector. Transformer off-load	6.4.103		Manufacturer to state
Switch disconnector. Line charging breaking current.	6.4.105		2A
Switch disconnector. Rated short circuit making current Dependant-manual Non dependant-manual	6.2		3kA/7.5kA (Peak) 10kA/25kA (Peak)
Mechanism type		Integral direct motor drive / Manual low level	Integral direct motor drive / Manual low level
Mech. box heater		110vAC	110vAC
No. of auxiliary contacts		8 NO, 8 NC	8 NO, 8 NC
Support insulators pollution class IEC 60815		Class III	Class III
Support insulators		Ceramic or composite	Ceramic or composite

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Ice load		10 mm	10 mm
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Appendix 2: Technical Declaration to be completed by the manufacturer

33kV Disconnecter

MANUFACTURER				
	ENATS 41-36 sub clause	Unit	33kV disconnecter	33kV switch-disconnector
Electrical Performance				
Type				
Number of poles				
Switch-disconnector category A-B	6.3			
Indoor, outdoor	1.2			
Rated voltage	1.4.1	kV		
Rated Frequency	1.4.3	Hz		
Rated insulation level (lightning impulse withstand voltage)	1.4.2	kV		
Rated insulation level (1 minute power RMS power frequency)	1.4.2	kV		
Disconnecter / Switch-disconnector rated normal current	1.4.4	A		
Rated short-time withstand current	1.4.5	kA		
Rated duration of short circuit	1.4.7	S		
Rated supply voltage of control, closing and opening devices	1.4.8	V		
Rated supply frequency of control, closing and opening and of auxiliary circuits.	1.4.9	ac/dc		
Switch disconnector. Rated mainly active load breaking current.	6.4.101	A	-	
Switch disconnector. Rated closed-loop breaking current	6.4.102	A	-	
Switch disconnector. Rated cable charging breaking current	6.4.104	A	-	
Switch disconnector. Transformer off-load	6.4.103	A	-	
Switch disconnector. Line charging breaking current.	6.4.105	A	-	
Switch disconnector. Rated short circuit making current	6.2	kA	-	
Dependant-manual				
Non dependant-manual				
Ice load		mm		
Support Insulators				
Type				
Pollution class				
Creepage distance		mm		
Maximum terminal loading		kG		

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Disconnecter Mechanism				
Manufacturer's Designation				
Mechanism type: Dependent manual/dependent power				
Rated supply voltage of closing and opening devices and control circuits		V		
Motor rating (W)		W		
Operating time of opening and closing (s)		s		
Motor protection				
Manual operation: method of interlock with motor mechanism				
No of auxiliary contacts				
Key Interlock type				
Key Interlock: Method of interlock with motor control circuits.				
Padlocking facilities				
Mechanism Degree of Protection				
Surface preparation				
Mechanism box heater voltage		V		
Disconnecter Weights				
Mass of disconnector (minus mechanism)		kG		
Mass of Mechanism		kG		
Disconnecter Drawings				
Disconnecter General Arrangement				
Operating mechanism layout				
Control Schematic				
Earth Switch				
Manufacturer				
Manufacturer's Designation				
Type (e.g. chopper, pantograph etc)				
Rated short-time withstand current		kA		
Rated duration of short circuit		S		
Earth Switch Mechanism				
Manufacturer's Designation				
Mechanism type: Dependent manual/dependent power				
No of auxiliary contacts				
Key Interlock type				
Padlocking facilities				
Mechanism Degree of Protection				
Surface preparation				
Earth Switch Weights				
Mass of earth (minus mechanism) (kg)		Kg		
Mass of Mechanism		Kg		

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Earth Switch Drawings				
Earth Switch General Arrangement				
Operating mechanism layout				

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Appendix 3: Addendum to supplier requirements

Project-specific installation and protection requirements will be provided by Primary Engineering Projects for inclusion in this appendix.